CS3MI3 Assignment 1

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Question 1

Let $t, t_0, t_1, t_2 \in \text{arith}$, then:

$$\begin{aligned}
\text{esses}(0) &= 0 \\
\text{esses}(\text{true}) &= 0 \\
\text{esses}(\text{false}) &= 1 \\
\text{esses}(\text{pred } t) &= \text{esses}(t) \\
\text{esses}(\text{succ } t) &= 1 + \text{esses}(t) \\
\text{esses}(\text{iszero } t) &= 1 + \text{esses}(t) \\
\text{esses}(\text{if } t_0 \text{ then } t_1 \text{ else } t_2) &= 1 + \text{esses}(t_0) + \text{esses}(t_1) + \text{esses}(t_2)
\end{aligned}$$

Question 2

Firstly, we will define $\text{size}(t),\ t, t_0, t_1, t_2 \in \text{arith}$:

$$\begin{aligned}
\text{size}(0) &= 1 \\
\text{size}(\text{false}) &= 1 \\
\text{size}(\text{true}) &= 1 \\
\text{size}(\text{succ } t) &= 1 + \text{size}(t) \\
\text{size}(\text{pred } t) &= 1 + \text{size}(t) \\
\text{size}(\text{iszero } t) &= 1 + \text{size}(t) \\
\text{size}(\text{if } t_0 \text{ then } t_1 \text{ else } t_2) &= 1 + \text{size}(t_0) + \text{size}(t_1) + \text{size}(t_2)
\end{aligned}$$

Next, we will prove that the property holds from our base cases:

Case $t = 0$

$$\begin{aligned}
\text{esses}(0) &\leq \text{size}(0) \\
0 &\leq 1
\end{aligned}$$

Case $t = \text{true}$

$$\begin{aligned}
\text{esses}(\text{true}) &\leq \text{size}(\text{true}) \\
0 &\leq 1
\end{aligned}$$

Case $t = \text{false}$

$$\begin{aligned}
\text{esses}(\text{false}) &\leq \text{size}(\text{false}) \\
1 &\leq 1
\end{aligned}$$

Therefore, the property holds for all bases cases. Now we will prove that the property holds for all inductive cases $t \in \text{arith}$:

Case $t = \text{succ } t_0$

Assuming:

$$\text{esses}(t_0) \leq \text{size}(t_0)$$

then,

$$\begin{aligned}\text{esses}(\text{succ } t_0) &\leq \text{size}(\text{succ } t_0)\\ 1 + \text{esses}(t_0) &\leq 1 + \text{size}(t_0)\\ \text{esses}(t_0) &\leq \text{size}(t_0)\end{aligned}$$

Using the **induction hypothesis**, our property holds.

Case $t = \text{pred } t_0$

Assuming:

$$\text{esses}(t_0) \leq \text{size}(t_0)$$

then,

$$\begin{aligned}\text{esses}(\text{pred } t_0) &\leq \text{size}(\text{pred } t_0)\\ \text{esses}(t_0) &\leq 1 + \text{size}(t_0)\end{aligned}$$

As $\text{esses}(t_0) \leq \text{size}(t_0)$ by our assumption, $\text{esses}(t_0) \leq 1 + \text{size}(t_0)$ will hold, trivially.

Case $t = \text{iszero } t_0$

Assuming:

$$\text{esses}(t_0) \leq \text{size}(t_0)$$

then,

$$\begin{aligned}\text{esses}(\text{iszero } t_0) &\leq \text{size}(\text{iszero } t_0)\\ 1 + \text{esses}(t_0) &\leq 1 + \text{size}(t_0)\\ \text{esses}(t_0) &\leq \text{size}(t_0)\end{aligned}$$

Using the **induction hypothesis**, our property holds.

Case $t = \text{if } t_0 \text{ then } t_1 \text{ else } t_2$

Assuming:

$$\begin{aligned}\text{esses}(t_0) &\leq \text{size}(t_0),\\ \text{esses}(t_1) &\leq \text{size}(t_1),\\ \text{esses}(t_2) &\leq \text{size}(t_2)\end{aligned}$$

then,

$$\begin{aligned}\text{esses}(\text{if } t_0 \text{ then } t_1 \text{ else } t_2) &\leq \text{size}(\text{if } t_0 \text{ then } t_1 \text{ else } t_2)\\ 1 + \text{esses}(t_0) + \text{esses}(t_1) + \text{esses}(t_2) &\leq 1 + \text{size}(t_0) + \text{size}(t_1) + \text{size}(t_2)\\ \text{esses}(t_0) + \text{esses}(t_1) + \text{esses}(t_2) &\leq \text{size}(t_0) + \text{size}(t_1) + \text{size}(t_2)\end{aligned}$$

Using the **induction hypothesis**, our property holds.

Question 3

```
esses :: ArithTerm -> Int
esses TmTrue = 0
esses TmFalse = 1
esses TmZero = 0
esses (TmSucc t) = 1 + esses t
esses (TmPred t) = esses t
esses (TmIsZero t) = 1 + esses t
esses (TmIfThenElse t0 t1 t2) = 1 + esses t0 + esses t1 + esses t2
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